

EXECUTIVE SUMMARY

Improving Term Deposit Conversion: A Data-Driven Telemarketing Strategy for a Retail Bank

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Business Problem

A retail bank invested substantial resources in telemarketing campaigns to promote term deposits but achieved a relatively low subscription rate of 11.27% across 41,188 customer contacts. Without understanding which customers are most likely to subscribe and which campaign practices are most effective, marketing resources risk being allocated inefficiently, leading to unnecessary costs and missed opportunities.

Business Objective

This project identifies which customer segments, contact strategies, and market conditions are associated with successful term deposit conversions and translates these findings into practical targeting and channel recommendations to improve future campaign efficiency.

Methodology

Business Understanding → Data Audit → Data Preparation → Exploratory Analysis → Insight Generation → Business Recommendations → Dashboard for Decision Support

Data preparation and exploratory analysis were conducted using SQL Server through six thematic analysis scripts covering:

- Campaign performance
- Customer segmentation
- Financial profile analysis
- Campaign strategy evaluation
- Macroeconomic indicators
- Multidimensional analysis

The findings were consolidated into an interactive Power BI dashboard designed to support campaign planning and decision-making.

Key Findings & Business Implications

- **Previous campaign response is the strongest conversion indicator**

Customers who subscribed in previous campaigns achieved a 65.11% conversion rate, nearly six times higher than the campaign average (11.27%). Customer history represents the most valuable existing signal for prioritizing future outreach.

- **Older customers represent the strongest demographic opportunity**

Retirees and customers aged 60+ achieved a 39.56% conversion rate, more than 3.5 times the campaign average, making them the strongest demographic segment identified.

- **Cellular outreach significantly outperforms landline**

Customers contacted through cellular channels achieved a 14.74% conversion rate, compared with 5.23% for landlines. This suggests that shifting outreach capacity toward mobile contacts could improve campaign efficiency.

- **Financial product ownership provides limited targeting value**

Housing loan and personal loan status showed limited predictive value. Removing these variables from targeting criteria can simplify segmentation while allowing analysts to focus on stronger indicators.

- **Interest-rate conditions show a potential timing effect**

Lower Euribor rates were associated with higher subscription rates. However, this relationship should currently be treated as a directional signal rather than a fixed campaign rule until additional month-level analysis confirms the effect.

Business Recommendations

1. Prioritize customers based on previous conversion history.

Build target lists by first identifying previous responders, then applying additional criteria such as age group and contact channel.

2. Adopt a cellular-first contact strategy.

Prioritize mobile outreach and reserve landline contacts for cases where no mobile number is available.

3. Optimize contact frequency.

Limit repeated attempts on low-response leads and redirect outreach capacity toward new prospects.

4. Remove low-value targeting variables.

Exclude housing and personal loan status from segmentation criteria due to their limited contribution.

5. Continue monitoring macroeconomic conditions.

Track the relationship between Euribor rates and campaign performance before incorporating interest-rate conditions into campaign scheduling.

Business Impact

The analysis identifies clear opportunities to improve campaign efficiency without increasing marketing expenditure. Prioritizing customers with previous positive responses, supported by demographic and channel-based targeting, provides a practical approach for allocating limited outreach capacity toward higher-probability leads.

At a Glance

Metric	Result
Records analyzed	41,188 customer contacts
Variables analyzed	21
Baseline conversion rate	11.27%
Highest-converting segment	Previous responders — 65.11%
Strongest demographic segment	Age 60+ / Retirees 39.56%
Best-performing channel	Cellular 14.74% vs. 5.23% landline
Tools	SQL Server (T-SQL), Power BI

Full analysis, SQL scripts, Power BI dashboard, and documentation:

<https://github.com/FARAHBENSALEM01/Bank-Marketing-Campaign-Analysis>